PRODUCT REQUIREMENTS DOCUMENT

Merchant Onboarding Recovery Agent

A deterministic, human-governed recovery workflow for stalled and rejected merchant applications

RAZORPAY | PRODUCT & ONBOARDING OPERATIONS

Review Draft 1.3 | 9 November 2026

Status: not approved for live merchant communication

INDEX

SECTION	NO.
Summary	01
Problem and user	02
Goals, non-goals and feature list	03
Scope	04
Inputs	05
Decision logic	06
Blank-code handling	07
Channel and language rules	08
Message rules	09
Guardrail checker	10
Human-in-the-loop	11
Follow-up sequence	12
Happy, negative, edge and data-anomaly cases	13
Outputs	14
Success criteria	15
Test plan	16
Risks and controls	17
Assumptions and open questions	18
Tools and AI in the build	19
Requirement status and release scope	20
System context and integration contracts	21
Case lifecycle and communication orchestration	22
AI classification, evaluation and model governance	23
Security, privacy and data governance	24
Operational ownership, SLAs and incident response	25
Product analytics and experiment design	26
Acceptance criteria and traceability	27
Production release gates	28
Decision and dependency register	29
Change log	30
Appendix A: Message library (English master)	A

1. Summary

This PRD specifies an agent that diagnoses 75 stalled or rejected merchant applications, identifies 67 as potentially recoverable in the synthetic data pack, and prepares checked nudges in six languages while routing risky cases to a person. The Rs 933.9 lakh figure is estimated monthly TPV associated with the potentially recoverable cohort; it is not recovered TPV or a forecast of realised recovery.

An agent that reads every stalled or rejected merchant application, works out why it stalled, and decides whether it can be recovered, must be closed, or goes to Compliance. For each recoverable merchant it builds a short fix message in the merchant's language from approved pieces, checks it against safety rules, and picks WhatsApp or email. Every case gets a colour: green sends automatically, amber waits for a person, red is owned by a person. It is built for Razorpay's Onboarding Ops team, and it also tells the product team which fixable causes block the most merchants.

- Result on the synthetic data pack: 67 of 75 in-scope applications are classified as potentially recoverable, with Rs 933.9 lakh in associated estimated monthly TPV. This is an opportunity estimate, not realised or forecast recovered TPV. Five cases are routed for closure approval and three to Compliance.
- **Top 3 systemic causes:** name mismatch on PAN or bank account (R01 and R03 grouped, since one fix covers both: 16 applications, Rs 132.2 lakh), missing website policy pages (7, Rs 128.1 lakh) and GST problems (6, Rs 106.3 lakh). Another 22 merchants stopped with no reason recorded (Rs 307.6 lakh). Section 14 shows the same ranking with the two name-mismatch codes split.
- **Safety:** all 75 drafted messages pass the guardrail checker, and the same input always gives the same output.

2. Problem and user

Onboarding Ops cannot efficiently diagnose and prepare accurate, safe recovery communications for 75 stalled or rejected applications across six languages by hand. The synthetic pack associates Rs 933.9 lakh of estimated monthly TPV with the 67 cases classified as potentially recoverable; this is an opportunity estimate, not a guarantee that these applications are one fix away from activation.

Why it hurts, in numbers

- 75 of 100 applications are stuck: 67 Stalled and 8 Rejected.
- The reason is unclear for many of them. 49 have a catalog code, 4 have only a short reviewer note (two in Hinglish), and 22 have no reason at all.
- 48 of the 67 recoverable merchants prefer a language other than English, and 18 use Tamil, Kannada or Marathi.
- Time works against recovery. Rs 505.9 lakh of the recoverable TPV has been idle for more than 14 days.
- People are scarce. A small team can make about 10 calls a week (assumed), against 26 merchants who are high value or struggling.

Who uses it

User	Role	What they need	What the agent gives them
Onboarding Ops agent or onboarding RM (primary)	Works the queue every day, calls merchants, approves closures	Who to act on first, what is wrong, what to say	A colour-coded work queue, a one-screen case card with the message and a call script
Sales RM or key account manager	Owns the largest merchants	A heads-up when a high-value merchant is stuck	The 8 value call slots, with the case card
Compliance team	Owns screening cases	No leak of reasons, control over contact	Fixed holding notes they approve once and can pause per case
Merchant (recipient)	Small business owner finishing sign-up	Plain words, in their language, telling them exactly what to fix	One short message with one task and one link
Product team	Fixes the onboarding flow	Which causes block the most merchants	A ranked cause report with counts and TPV

The operator, not the merchant, is the main user. The merchant is who the operator serves.

3. Goals, non-goals and feature list

The agent's job is to get every recoverable merchant one correct, safe message, and to put human time only where it changes the outcome.

Goals

1. Diagnose every in-scope application, including those with a blank code or no reason.
2. Give every recoverable merchant a correct fix message in their language, on a channel they agreed to.
3. Route every closure and compliance case to a person, with no fix-it message.
4. Spend the weekly call budget where a person adds the most value.
5. Tell the product team which fixable causes block the most merchants.

Non-goals

- It does not decide compliance outcomes or explain them.
- It does not close any application without a person's approval.
- It does not let AI write free text. Every message is built from approved pieces.
- It does not promise approval, dates, refunds, waivers or speed.
- It does not predict who will resume. The pack has no outcome data, so no recovery rate is claimed.
- It does not contact real merchants. The data is synthetic and this build is a dry run.

Feature list: must-haves first, then value-adds

Built = runs in the current agent. Designed = specified here, not yet built in the app. Proposed = next step.

Feature	Why	Type	Status
Diagnose every non-activated application, blank codes included	Brief minimum	Must-have	Built
Fixable and recoverable decision per application	Brief minimum	Must-have	Built
Nudge in the preferred language; WhatsApp only if opted in	Brief minimum	Must-have	Built; all six languages approved
No workaround for restricted categories; no compliance reason revealed	Brief minimum	Must-have	Built and checked
Summary: recoverable count, TPV at stake, top 3 causes	Brief minimum	Must-have	Built
Cause ranking for the product team	Brief task	Must-have	Built
Priority by TPV and days idle	Brief stretch	Stretch	Built
Follow-up sequence with stop rules	Brief stretch	Stretch	Built (drafts); sending designed
Product change for the top cause	Brief stretch	Stretch	Proposed (first fix named in section 14.5)
Green, amber, red light on every case, with a loud run report	Nothing hides in code	Value-add	Built
RM case card with message, English meaning and call script	One screen to act	Value-add	Built (Applications tab, steps 1 to 4)

Feature	Why	Type	Status
Call budget: 8 value slots plus 2 stuck small-merchant slots	Human time where automation failed	Value-add	Built
Safety override: risky reviewer text beats a fixable code	A blocked merchant never gets a cheerful nudge	Value-add	Built
AI second opinion on blank-code text, with an 80% confidence floor	AI in one narrow, checked place	Value-add	Built (saved answers, so every run is identical)
English meaning beside every non-English message	Anyone can check any language	Value-add	Built
Guardrail tester: paste a message, see which rule it breaks	Shows the safety net working	Value-add	Built (Safety and tests tab)
Decision receipt per case: rule trace, library version, run hash	Audit and reproducibility	Value-add	Partly built (rule trace on every case card, plus the run hash)
Data-quality flags, such as idle longer than account age	Honest about bad data	Value-add	Built
Fairness view: who the ranking crowds out	Bias made visible	Value-add	Designed (not in the current app)
Kill switch for all automated sends	One click stops everything	Value-add	Designed
Time limit on amber cases	Nothing waits forever	Value-add	Designed
One-tap "What stopped you?" for abandoned merchants	Turns 22 unknowns into data	Value-add	Proposed
Reply sorting: help, done, angry, opt-out, asks why	Faster human follow-up	Value-add	Partly built (HELP, STOP and any other reply are routed)
Deep link to the exact field to fix	Fewer taps to finish	Value-add	Proposed

4. Scope

In scope are the 75 applications that are Stalled or Rejected; the 25 Activated ones are out of scope but stay visible as "Excluded", so the output covers all 100 rows.

Group	Applications	Treatment
Stalled	67	Diagnosed. Recoverable ones enter the nudge sequence
Rejected	8	Diagnosed. Never enter the automated sequence. All 8 are R10 or R12
Activated	25	Excluded. No code, no text, no message
Total	100	

- **Data:** the synthetic Data Pack only, tabs P3_Applications (100 rows, 15 columns) and P3_RejectionCatalog (13 codes). Snapshot date 30 Sep 2026.
- **Policy source of truth:** the catalog's meaning, fixable flag and standard fix, followed as given even where real practice may differ.
- **Languages:** English, Hinglish, Hindi, Marathi, Tamil and Kannada.
- **Channels:** WhatsApp (opted-in merchants only) and email.
- **Out of scope for this build:** live sending, real merchant data, merchant names or contact details, and any change to Razorpay's onboarding policy.

5. Inputs

The agent reads 11 of the 15 columns to decide and write. The other 4 are used only for reporting and data checks, so a merchant's city or business type can never change their treatment.

Order	Column	What the agent does with it	What it changes
1. Gate	application_id	Keys every output row and breaks ranking ties, so results repeat exactly	Reproducibility
1. Gate	application_status	Activated is excluded. Rejected is locked out of the automated sequence. Stalled continues	In or out of scope
2. Reason	rejection_code	Looked up in the catalog (R01 to R12)	Cause, fixable, standard fix
2. Reason	rejection_message	Confirms the code, infers a code when it is blank, and can override a fixable code if it signals screening or a restricted category	Cause, safety override
2. Reason	step_reached	Names the step in abandoned-case messages and sanity-checks the code against its step	Message wording, data flag
3. Routing	expected_monthly_tpv_inr_lakh	30 or more puts a merchant in Lane A, and it drives the priority score	Lane, call priority
3. Routing	attempts	3 or more means the merchant is struggling	Lane B, empathy line
3. Routing	support_contacts	2 or more means the merchant is struggling	Lane B, empathy line
3. Routing	days_since_last_activity	Sets the tone (7 or fewer, 8 to 30, 31 or more) and the freshness factor	Tone, sequence length, priority
4. Message	preferred_language	Picks the language and script	Message
4. Message	whatsapp_opted_in	Yes allows WhatsApp. No means email	Channel
Reporting	business_type, industry, city	Fairness check and summary only	Nothing
Reporting	signup_date	Data check: idle days cannot exceed account age	Data flag only

What is missing from the data, and what we do

- No phone number or email address. We assume the contact used at sign-up exists. In the demo, any test send goes only to the builder's own inbox.
- No merchant or business name. The greeting is generic: "Dear Valued Merchant".
- No resume link. Every message carries a [resume link] placeholder, assumed unique per application.
- No outcome data (who resumed). No recovery rate is claimed.

Input checks run before anything else

Check	Result on the pack	If it fails in future data
Duplicate application_id	0	Run stops with an error. Nothing is drafted
Missing status, language, opt-in, TPV or idle days	0	Missing opt-in is treated as No (email). Missing language falls back to English with an amber flag. Missing TPV or idle days: no call slot, amber flag
Language outside the six supported	0	English draft, amber flag
Code not in the catalog	0	Treated as unclassified: no message, a person decides
Code at an unexpected step	0 of 45 fixable codes	Data flag, decision unchanged
Idle days longer than account age (snapshot minus signup date)	21 rows, for example A3016: signed up 22 Sep, shown 37 days idle on 30 Sep	Flagged "Date check" in the output. Decisions unchanged. Raised with the data owner

The 21 date anomalies are reported, not fixed. The agent trusts days_since_last_activity because it is the field the brief asks us to prioritise on, and changing it would be inventing data.

6. Decision logic

Rules, not AI, decide every route. The code sets the cause, the cause sets fixable and recoverable, and the lane and call budget only decide how much human help a recoverable merchant gets.

Code to action

Code	Cause in plain words	Fixable	Recoverable	In pack	Action	Owner	Light
R01	PAN name differs from business name	Yes	Yes	9 (1 inferred)	Nudge: upload the matching PAN or correct the business name	Automated; call if slotted	Green or amber
R02	GST certificate invalid or mismatched	Yes	Yes	6	Nudge: correct GSTIN or upload a current certificate	Automated; call if slotted	Green or amber
R03	Bank account name mismatch	Yes	Yes	7	Nudge: use an account in the business name, or upload a bank letter	Automated; call if slotted	Green or amber
R04	Bank verification failed	Yes	Yes	4	Nudge: re-enter account number and IFSC, check the account is active	Automated; call if slotted	Green or amber
R05	Document unreadable	Yes	Yes	4 (1 inferred)	Nudge: upload a clear full-page scan under 5 MB	Automated; call if slotted	Green or amber
R06	Website missing policy pages	Yes	Yes	7	Nudge: add Terms, Privacy, Refund and Contact pages, resubmit	Automated; call if slotted	Green or amber
R07	Website not live	Yes	Yes	3	Nudge: make the site public with products, prices, contact	Automated; call if slotted	Green or amber
R08	Address proof missing or outdated	Yes	Yes	1	Nudge: upload a recent proof matching the address	Automated; call if slotted	Green or amber
R09	Signatory details incomplete	Yes	Yes	3 (1 inferred)	Nudge: upload a valid signatory ID with matching name and date of birth	Automated; call if slotted	Green or amber
R10	Restricted business category	No	No	5 (1 inferred)	Polite closure. No workaround, no appeal path, no link	A person approves each	Red

Code	Cause in plain words	Fixable	Recoverable	In pack	Action	Owner	Light
R11	Declared category differs from website	Yes	Yes	1	Nudge: select the correct category or share a note with evidence. Never say which category to pick	Automated; call if slotted	Green or amber
R12	Compliance or sanctions screening hit	Needs compliance review	No, until Compliance clears it	3	Fixed holding note on day 0 and day 7. No reason, no timeline, no link	Compliance	Red
R13	Abandoned, no rejection recorded	Yes	Yes	22	Nudge naming the step where they stopped. Never guesses a reason	Automated; call if slotted	Green or amber

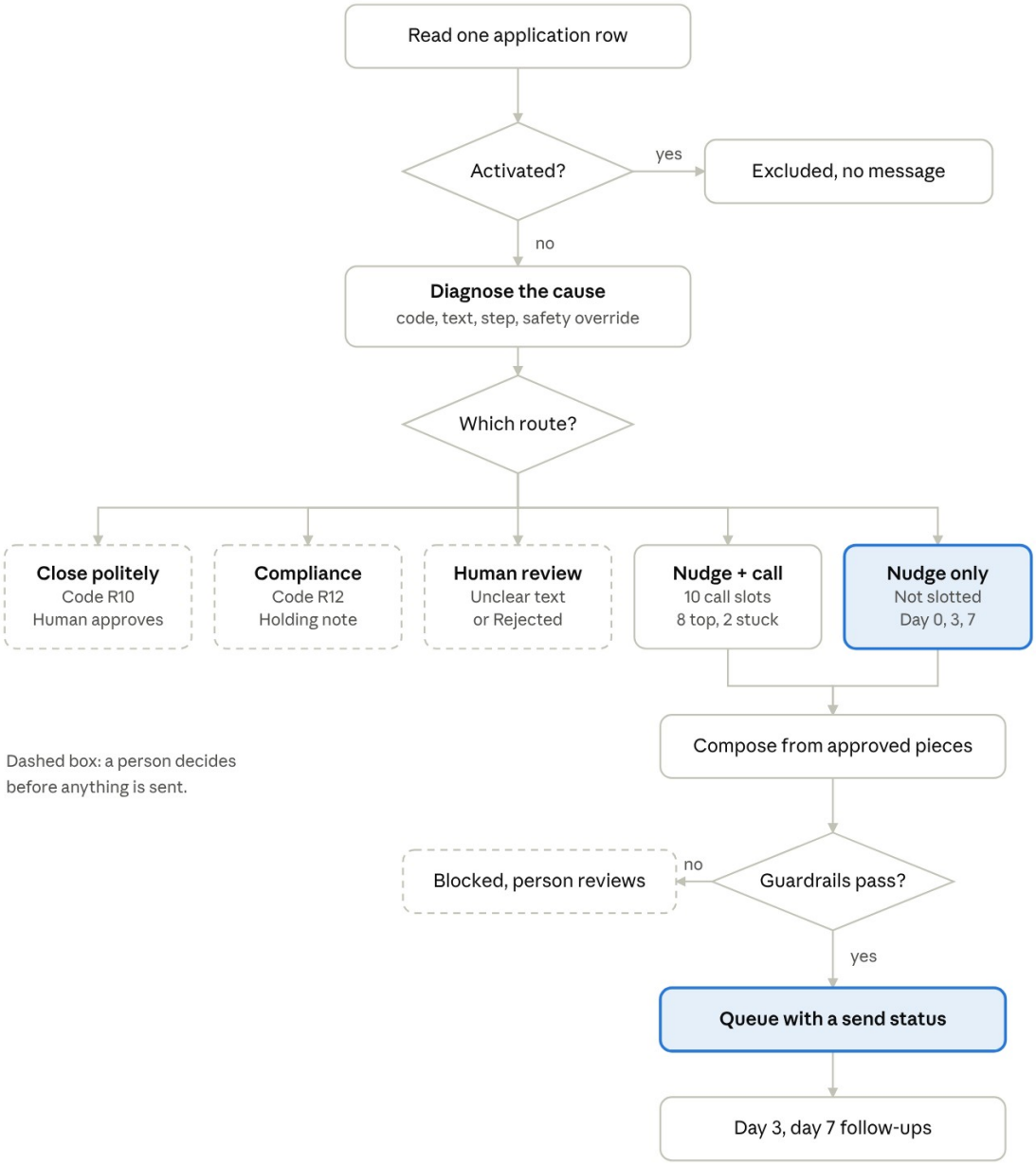
Fixable versus recoverable. Fixable comes from the catalog: 67 Yes, 5 No, 3 "needs compliance review". Recoverable is our decision: 67 Yes, 8 No. They match here because no fixable case is blocked by a Rejected status. R12 counts as not recoverable until Compliance clears it, and a cleared case re-enters the pipeline from the start.

Routing order (first match wins)

Step	Condition	Route	Fires in pack
1	Status is Activated	Excluded, no message	25
2	Code is R10, or the text signals a restricted category (safety override)	Polite closure after a person approves	5
3	Code is R12, or the text signals screening (safety override)	Compliance holding note	3
4	Text could not be classified, or AI and rules disagree (section 7)	Person decides, no message	0
5	Status is Rejected but the code looks fixable	Person decides, no automated message	0
6	Everything else	Recoverable: lane, call budget, nudge sequence	67

Steps 4 and 5 never fire on this pack, but they exist because a cheerful nudge to a merchant who is really blocked would break a hard rule.

Five routes, and unsafe cases never reach a nudge



application flow · 5 routes, 3 gates

Read top to bottom: activated merchants drop out, the rest are diagnosed, each takes one of five routes, and only the two nudge routes reach the message composer and the guardrail check. Dashed boxes are cases a person owns.

Lanes: what kind of help a recoverable merchant gets

Lane	Rule	Merchants	TPV (Rs lakh)	Help
A: White-glove	Expected TPV 30 lakh or more	7	328.9	Written nudge on day 0; usually in the top 8 for a call (all 7 are today); a named owner

Lane	Rule	Merchants	TPV (Rs lakh)	Help
B: Assisted	3 or more attempts, or 2 or more support contacts	19	148.6	Nudge with the empathy line and a help offer; a call if slotted or if they reply
C: Self-serve	Everyone else	41	456.4	Automated day 0, 3 and 7 sequence

A lane says what kind of help fits. A separate "Call this week" flag says who actually gets a call, because calls are limited.

Call budget: who gets a call this week

The weekly budget is a setting the ops lead changes. The default is 10 calls.

- **8 value slots:** the top 8 recoverable merchants by priority score, from any lane. Today: all 7 Lane A merchants plus A3068 (Rs 25.6 lakh, idle 4 days, quick bank fix).
- **2 stuck small-merchant slots:** the best-scoring Lane B merchants below the median TPV of Rs 8.0 lakh. Today: A3009 (Tamil, GST, 4 attempts) and A3020 (Kannada, policy pages, 4 attempts). Automation has already failed these merchants, so a person adds the most here. They also form a test group: compare them with the 11 similar stuck small merchants who were not called.
- **Next in line (5):** A3054, A3050, A3091, A3092 and A3008, called in that order if capacity frees up.
- **Inferred-code rule:** A3081 and A3022 hold value slots but their codes were read from text. A person confirms the code the same day, or the slot passes to the next in line.

Option	Calls	TPV covered (Rs lakh)	Share of recoverable TPV
Call every Lane A and B merchant	26	477.5	51.1%
Top 10 by score only	10	407.8	43.7%
Chosen: 8 by score plus 2 stuck small merchants	10	366.5	39.2%
Lane A only	7	328.9	35.2%

The chosen rule needs 16 fewer calls than calling all of Lanes A and B. It gives up 4.5 points of coverage against a pure top 10, and that is the price of putting two calls where automation has already failed.

Priority score: the order for calls

The score is risk-adjusted TPV: the money at stake, discounted for how likely the merchant is to come back soon. It orders calls only. Every recoverable merchant gets the automated sequence regardless of score.

Priority score = TPV × Freshness × Ease × Evidence

Factor	Values	Why
TPV	Expected monthly TPV, Rs lakh	Money at stake
Freshness	1.0 if idle 7 days or fewer, 0.85 for 8 to 14, 0.6 for 15 to 30, 0.35 for 31 or more	Recent merchants are more likely to respond
Ease	1.0 if the catalog fix takes 1 day, 0.9 for 2 days, 0.8 for 3 days	Quick fixes go live sooner
Evidence	1.0 catalog code, 0.9 inferred from text, 0.7 abandoned with no reason	Less certainty about the cause ranks lower

Formula check: does it make sense?

- **Each factor answers one business question:** how much, how recent, how hard and how sure.
- **Robust to the weights:** with softer freshness weights (1.0, 0.9, 0.75, 0.5) the top 10 is identical, 10 of 10.
- **Mostly follows TPV, by design:** a plain TPV sort shares 8 of the top 10. The other factors push down stale merchants. Example: A3023 has the largest TPV (Rs 59.9 lakh) but has been idle 35 days, so it ranks 8th.
- **Attempts and support contacts are left out on purpose:** they signal both effort and frustration, so their direction is unclear. Lane B and the stuck small-merchant slots handle them instead.
- **Cliffs at the band edges are accepted:** A3088 sits at exactly 14 days. A cliff only changes the call order, never whether a merchant is messaged.
- **Ties** are broken by application ID, so the order is identical on every run.
- **The weights are assumptions,** to be refitted on real resume rates after the first weeks live.

Fairness: who the ranking crowds out

- Proprietorships are 29 of the 67 recoverable merchants (43%) but only Rs 132.0 lakh of TPV (14%), averaging Rs 4.6 lakh. A pure score ranking gives them 0 of 10 calls. The stuck small-merchant slots give them 2 of 10.
- Regional-language merchants are not shortchanged: 18 of 67 (27%), with 3 of the top 10 places.
- Seven merchants hold 35% of recoverable TPV, so the ranking is concentrated by design.
- **Stance:** ranking by money is a value choice. We use it only to ration calls, we show who it crowds out, and we reserve slots for them.

7. Blank-code handling

A blank code is never guessed silently. Keyword rules and an AI second opinion must agree, a person confirms, and anything unclear gets no message.

What blank looks like in the pack

Situation	Applications	Treatment
Activated, no code, no text	25	Expected. Excluded
No code, no text, not activated	22	R13 abandoned. The message names the step (Business details 10, KYC docs 6, Bank verification 6) and never guesses a reason
No code, but reviewer text present	4	Code inferred from the text, then confirmed by a person (below)

The 4 inferred cases: A3022 "Doc blurred and cropped" becomes R05, A3028 "PAN ka naam ... match nahi kar raha" becomes R01, A3081 "Signatory ka ID proof upload nahi hua" becomes R09, and A3051 "Business category not supported by policy" becomes R10.

How a code is inferred

1. **Keyword rules first.** The text is matched against a fixed keyword list per code, in English and Hinglish. Risk words are checked before anything else: "screening" or "compliance" means R12, "restricted" or "not supported" means R10.
2. **AI second opinion.** An AI model reads the same text and must answer with one catalog code (R01 to R12) or UNCLEAR, a confidence from 0 to 100, and the exact words it relied on. It cannot write a message or choose a route.
3. **Combine.**
    - Rules and AI agree, and AI confidence is 80 or more: the code is used, marked "Inferred, confirm" (amber). The draft is prepared, but nothing is sent until a person confirms.
    - They disagree, AI says UNCLEAR, or confidence is under 80: no draft. The case is marked unclassified (red) and a person reads it.
    - Either one says R10 or R12: the case goes to the closure or Compliance route and a person decides. Safety wins over agreement.
4. **Reproducible.** AI answers are saved with the input text. A rerun reads the saved answer instead of asking again, so the output never changes between runs.

Why 80%. A wrong code sends the wrong fix to a real merchant, which wastes their effort and our credibility. The blank-code volume is tiny (4 of 75), so sending a borderline case to a person costs little. 80% leaves a clear margin above a coin flip between two plausible codes. If the gold set shows any wrong answer at 80% or more, the floor rises.

Safety override. Even when a code is present, text that signals screening or a restricted category wins over a fixable code. A merchant who is really blocked never gets a cheerful fix-it nudge.

Status today. The keyword rules are built and infer a code for all 4 cases, each consistent with its reviewer text; whether they match the hidden answers is tested by the gold set. The AI second opinion is built: its answers were produced once, saved, and are read on every run, so results never change.

8. Channel and language rules

Each merchant gets one channel and one language: WhatsApp only with opt-in, otherwise email, always in their preferred language.

Channel

Rule	Detail	In scope
WhatsApp	Only if whatsapp_opted_in is Yes. Short form, ends with Reply "STOP" to stop these messages. Hard limit 1,000 characters	56
Email	Everyone not opted in. Same body plus a subject line and the sign-off "Razorpay Onboarding Team"	19
One channel per merchant	Never both, never two messages on the same day	All
Sending window	9am to 7pm India time only	All
WhatsApp send fails	Falls back to email after a person checks; never a second WhatsApp	Designed

In production, outbound WhatsApp needs pre-approved business templates. Our fixed pieces map one-to-one to templates with variables (step, short issue, link), so approval is a one-time task.

Language

Language	Script	In scope	Greeting	Review status
English	Latin	21	Dear Valued Merchant,	Approved by owner
Hinglish	Latin (Roman Hindi)	18	Dear Valued Merchant,	Owner to confirm
Hindi	Devanagari	14	आदरणीय व्यापारी,	Owner to confirm
Tamil	Tamil	13	மதிப்பிற்குரிய வணிகரே,	Approved by native speakers
Kannada	Kannada	5	ಗೌರವಾನ್ವಿತ ವ್ಯಾ ಪಾರಿಗಳೇ,	Approved by native speakers
Marathi	Devanagari	4	आदरणीय व्यापारी,	Approved by native speakers

- **Preferred language always.** The message goes out in preferred_language. A missing or unsupported language falls back to English with an amber flag.
- **Regional wording.** Tamil, Kannada and Marathi messages were reviewed and approved by native speakers, so they send like any other language, with spot checks.
- **English meaning beside every message.** A reviewer who does not read the language can still check what it says.
- **Script check.** A message in the wrong script for its language is blocked. "Razorpay" stays in Latin letters inside every script.
- **Length.** The longest Day 0 message is 704 characters in English and 823 in Tamil (worst cases, v2 format). All are under the 1,000 limit.

9. Message rules

A good message names the exact problem, gives the exact fix, says why it matters, and offers one link, in warm and plain words. The AI never writes free text: every message is assembled from approved pieces (Appendix A).

Anatomy of a Day 0 message

1. Greeting: "Dear Valued Merchant," (translated per language), or "Dear [name]," when the data has a merchant name that passes the safety checks.
2. Opener, set by days idle (table below).
3. Steps-left line: "You are just three steps away from completing it." Counted from the current step to the end, including the current step. Four left reads "just a few steps". The four steps are Business details, KYC documents, Bank verification and Website/App review.
4. Empathy line, only for stuck merchants: "We are sorry this has taken some effort."
5. Problem line from the catalog, in plain words, starting "We noticed that..." or "We could not...", with the key clause in bold. Abandoned cases: "We noticed you paused at the {step} step."
6. Fix as bullets under "Here's how you can fix it:", or "(choose one)" where the fix has real alternatives (R01, R02, R03, R11). Abandoned cases use "Here are your next steps:". The key action is in bold.
7. Why it matters: "This is needed before you can start accepting payments." For abandoned cases: "Finishing your setup is needed before you can start accepting payments."
8. Link line: "Continue here: [resume link]".
9. Help line: "For assistance, reply "HELP" to this message. Our team will be happy to help you." A "HELP" reply goes to a person (section 11). Email says "reply to this email".
10. WhatsApp ends with "Reply "STOP" to stop these messages." Email adds a subject line and "Razorpay Onboarding Team".

Days idle	Tone	Opener
7 or fewer	Soft	Thank you for trusting us and starting your setup with Razorpay.
8 to 30	Reminder	Your Razorpay setup is still waiting for you.
31 or more	Gentle	We are here whenever you are ready to finish your Razorpay setup.

Good example (R01, idle 4 days, WhatsApp, English)

> *Dear Valued Merchant,*
>
> *Thank you for trusting us and starting your setup with Razorpay. You are just three steps away from completing it.*
>
> *We noticed that* ***the name on your PAN card does not match the business name in your application.***
>
> *Here's how you can fix it (choose one):*
>
> - ***Upload the PAN of the business*** *(or of the owner, for a proprietorship) that matches your application*
> - ***Or correct the business name*** *in your application to match your PAN*
>
> *This is needed before you can start accepting payments.*
>
> *Continue here: [resume link]*
>
> *For assistance, reply "HELP" to this message. Our team will be happy to help you.*
>
> *Reply "STOP" to stop these messages.*

Tone rules (from the owner's tone-of-voice guidelines)

- Human, warm and professional. Simple everyday words, short sentences.

- Every message covers context (where the application stands), relevance (why it matters) and action (what to do).
- No blame, no slang, no exclamation marks, no emojis, no pressure words such as urgent, immediately or last chance.
- Tone adapts: soft when fresh, reminder when waiting, gentle when long idle, empathy when stuck.
- One task and one link per message. Help is offered as a reply, not a second task.
- No numbers or dates except "5 MB" in the document fix. Steps are written as words. No idioms: short, literal sentences that translate cleanly.

Banned, and what we say instead

Banned example	Why	What we say instead
"We will approve you in 2 days."	A promise and a date	"This is needed before you can start accepting payments."
"Your business was flagged by our screening."	Reveals a compliance reason	"Your application is currently under review."
"Try applying under a different category." (to R10)	A workaround for a restricted business	Polite closure with no next step
"There is an issue with your documents."	Vague, no action	Name the exact document and fix
"You uploaded the wrong PAN."	Blames the merchant	"The name on your PAN card does not match..."
"URGENT: last chance to finish!"	Pressure, exclamation	The gentle headline
"Your account will go live very soon."	A vague promise	Never written: no free text is allowed
"You stopped because your bank details failed." (to R13)	Guesses a reason	"You stopped at the Bank verification step."

Length. English Day 0 messages run 82 to 123 words (median 106) and 482 to 704 characters, longer than a short-message rule of thumb because each must carry context, relevance and action. Follow-ups run about 50 words. The 1,000-character WhatsApp limit is our own choice (WhatsApp template bodies allow about 1,024). Every message in all six languages stays under it; the longest is 823 characters (Tamil).

Brand check. All approved English wording follows the tone guidelines. One deliberate exception: the owner kept "Dear Valued Merchant", although the guidelines list "Dear Valued Customer" as a generic opening to avoid. The greeting uses a merchant name ("Dear [name],") whenever the data has one that passes the safety checks. The P3 data has no name column, so today everyone gets "Dear Valued Merchant". Two further owner choices: no emojis, and no idioms, so the messages translate cleanly.

10. Guardrail checker

Every message, including closures and holding notes, passes the checker before it can be queued. A message that fails any rule is blocked, turns red and goes to a person; it is never sent with the problem.

#	Rule	How it is enforced	On failure
1	Never nudge R10, R12, unclassified or Rejected	Routing lock plus a code check inside the checker	Blocked, red
2	No workaround for a restricted category	Closure text checked for reapply, resubmit, alternative, workaround, change your, appeal, instead; "again" is checked in all six languages	Blocked, red
3	No compliance or screening reason	Screening, compliance, sanction, risk, restricted blocked in English and in Devanagari, Tamil and Kannada script (regional word lists)	Blocked, red
4	No invented date, amount, deadline or timeline	No digits allowed except "5 MB" and the link placeholder	Blocked, red
5	No promise of approval, refund, waiver or speed	Approved, approval, guarantee, within, today, tomorrow, will be live, refund of, waive, waiver, aaj, kal and native-script equivalents blocked. "Refund" alone is allowed because the R06 fix names the Refund/Cancellation Policy page	Blocked, red
6	WhatsApp only with opt-in, and short	Opt-in check, 1,000-character limit, STOP and HELP lines present, no emoji on every WhatsApp nudge	Blocked; falls back to email after a person checks
7	Right script for the language	Share of characters in the expected script	Blocked, red
8	Nudges carry the link; closures and holds do not	Presence check by message type	Blocked, red
9	No unfilled placeholders	Check for stray braces	Blocked, red
10	Abandoned cases never state a reason	R13 text checked for reason words (reject, mismatch, invalid, wrong, fail)	Blocked, red
11	Irreversible actions need a person	Closures and inferred R10 cannot be queued without approval	Held until approved
12	Same input, same output	Fixed rules, saved AI answers, ties broken by ID, output hash compared	Run flagged, nothing sent
13	Synthetic data only	No real merchant or personal data in the build	Not applicable

Results today

- All 75 drafted rows pass, including their Day 3 and Day 7 follow-ups.
- 1,272 message variants (every code, tone, channel, empathy state and follow-up, in all six languages) pass with 0 failures.
- 37 classification and adversarial tests pass, such as a WhatsApp nudge missing its STOP line, a fixable code with screening text, Hinglish text with no code, and unrecognised text.

- The app has a tester screen: paste any message and see which rule it breaks. Eight ready-made bad examples are included.

Known gap, stated openly. A keyword checker misses vague promises. In testing, "your account will go live very soon" passed. That is why no message contains free text, and why people review the approved library before it is used.

11. Human-in-the-loop

Every in-scope application gets exactly one light. Green sends on its own, amber waits for a person to check something, and red is owned by a person from start to finish. Today that is 56 green, 11 amber and 8 red.

Light	Send status	Count	Who acts	What they do	Time limit (proposed)
Green	Ready to send	56	Nobody, after a one-time sample	Sends in the window after the guardrail check. The ops lead samples the first batch of each code and language before auto-send is switched on	None
Amber	Call first, nudge is backup	8	Onboarding Ops or the RM	Calls the merchant. The Day 0 message still goes, so the fix is in writing. Pauses Day 3 and 7 if the call connects	Within the week
Amber	Confirm inferred code first	3	Onboarding Ops (triage)	Confirms the code read from reviewer text, or corrects it	Same working day; otherwise the call slot passes on
Red	Human only: approve the closure	5	Onboarding Ops lead	Approves each R10 closure, then sends it	2 working days
Red	Compliance holding note	3	Compliance	Approves the note wording once, can pause any case, answers any reply	Wording before launch
Red	Blocked by the checker, or unclassified	0	Onboarding Ops	Reads the case and decides	Same working day
Grey	Excluded (Activated)	25	Nobody	Nothing	None

An amber case not cleared within its time limit escalates to the ops lead, so nothing waits forever. Each case shows one reason, so the amber counts add up to 11. Ten calls are planned in all (8 value slots and 2 stuck small-merchant slots); 2 of those calls are inferred-code cases, so they sit under "Confirm inferred code first" and 8 sit under "Call first".

Why value alone does not need approval. A Lane A message uses the same approved pieces as a Lane C message, so a larger merchant does not make the text riskier. Approval is triggered by risk (closure, compliance, uncertain code, unreviewed language), not by size. Size buys a phone call instead.

Controls a person always has

- **Kill switch:** one setting pauses every automated send, for example during an incident.
- **Pause per case:** any message sequence can be paused from the case card.
- **Edit from the library only:** a person can swap pieces but cannot type free text into a merchant message.
- **Override with a reason:** a person can change a code or route; the reason is logged and feeds the weekly rule review.

Ownership

- **Onboarding Ops and RMs:** work the queue, make calls, confirm inferred codes, approve closures.

- **Compliance:** owns every R12 case and the holding-note wording. Support never explains reasons.
- **Native-speaker reviewers:** signed off the regional library; later batches are spot-checked.
- **Product team:** owns the rules, the library, the weekly cause report and this document.

12. Follow-up sequence

Each recoverable merchant gets at most three messages, on day 0, day 3 and day 7, and the sequence stops the moment the merchant replies, acts or opts out. The 13 merchants idle 31 days or more get one gentle message only.

Fix-it sequence (recoverable merchants)

Message	When	Sent only if	Wording (English)
1. First nudge	Day 0	In scope and recoverable	The full message from section 9
2. Reminder	Day 3	No reply, no new activity, idle 30 days or fewer at start	Greeting, then: "A reminder from Razorpay: your setup is still waiting on {short issue}. You are just [steps] away from completing it." Then the link, help and STOP lines.
3. Offer of help	Day 7	Still no reply and no new activity	Greeting, then: "Do you need help with {short issue}? You are just [steps] away from completing your Razorpay setup." Then the link, help and STOP lines.

{short issue} is a fixed label per code, for example "the PAN name match" or "your bank account verification". For abandoned cases the Day 3 reminder says "is waiting for you at the {step} step" and Day 7 asks "Do you need help with the {step} step?". Both follow-ups carry the steps-left line and the same HELP and STOP lines as Day 0. Email versions use the subjects "Your Razorpay setup: here is how to continue", "A reminder about your Razorpay setup" and "Need help with your Razorpay setup?".

Holding notes (R12, Compliance review)

Note	When	Wording (English)
1	Day 0	Dear Valued Merchant, thank you for applying to Razorpay. Your application is currently under review, and we have not forgotten about you. No action is needed from you at this time.
2	Day 7	Dear Valued Merchant, thank you for your patience. Your Razorpay application is still under review. No action is needed from you at this time.

No reason, no timeline, no link. After note 2, Compliance owns any further contact. A merchant under review who hears nothing feels forgotten: the 3 R12 merchants have been idle 23 to 33 days, and one is worth Rs 48 lakh.

Stop and pause rules

Trigger	What happens
Merchant replies, in any tone or language	Sequence stops. The thread goes to Onboarding Ops with no automatic answer. A reply of "HELP" is marked high priority and a person answers it
New activity on the application	Sequence pauses and the idle clock restarts
Application activated	Sequence stops. The next run excludes it
Merchant sends STOP or asks not to be contacted	Sequence stops and the opt-out is recorded
Case moves to closure or Compliance	All automation stops; a person owns the thread
A call connects and the merchant is working on it	The caller pauses messages 2 and 3

Trigger	What happens
Reviewer text changes to a screening or restricted signal	Safety override: sequence cancelled, case re-routed
Kill switch on	Every automated send pauses
After message 3 with no response	Logged as not recovered this cycle. Lane A and B go back to a person; Lane C feeds the weekly learnings

Guards on every send: one active sequence per application, one channel per merchant, never two messages on the same day, and only between 9am and 7pm India time.

Abandoned cases (proposed). Message 1 adds a one-tap question, "What stopped you?", with buttons for Documents, Bank details, Website and Something else. The answer becomes the cause on the next run, so the 22 unknowns shrink over time.

13. Happy, negative, edge and data-anomaly cases

Every situation below has a defined behaviour and a way to test it. Rows marked "Pack" use a real row from the data; rows marked "Injected" use a test input, because the pack has no such case.

#	Type	Scenario	Expected behaviour	How to test	Result
1	Happy	R01, Hindi, fresh, small (A3004)	Hindi PAN fix, soft headline, WhatsApp, green	Pack	Pass
2	Happy	R02, Marathi, idle 38 days, struggling (A3001)	Marathi GST fix, gentle headline, empathy line, Day 0 only, sends automatically	Pack	Pass
3	Happy	R13 abandoned, no reason (A3005)	Names the KYC documents step, no reason guessed, green	Pack	Pass
4	Happy	Not opted in to WhatsApp (A3068)	Email with subject line and sign-off, no STOP line	Pack	Pass
5	Happy	Highest TPV, idle 35 days (A3023)	Value call slot, gentle Kannada message, Day 0 only	Pack	Pass
6	Negative	R10 restricted category (A3039)	Polite closure, no workaround, no link, red until a person approves	Pack	Pass
7	Negative	R12 screening hit (A3094)	Holding note, no reason, no link, red, Compliance owns	Pack	Pass
8	Negative	Blank code, text says category not supported (A3051)	Inferred R10, closure route, a person approves	Pack	Pass
9	Negative	Fixable code, but text mentions screening	Safety override to the Compliance route, no nudge	Injected	Pass
10	Negative	Status Rejected with a fixable code	Person decides, no automated message	Injected	Pass after a fix (test found a bug)
11	Negative	Reviewer text matches nothing known	Unclassified, red, no message	Injected	Pass
12	Negative	Message contains "approved within 2 days"	Blocked by rules 4 and 5, red	Injected into the tester	Pass
13	Negative	WhatsApp chosen for a merchant not opted in	Blocked by rule 6, falls back to email	Injected	Pass
14	Edge	Blank code, Hinglish text (A3028)	R01 inferred, amber until confirmed	Pack	Pass
15	Edge	Inferred code on a value call slot (A3081, A3022)	Confirm the same day, or the slot passes to the next in line	Pack	Designed
16	Edge	Merchant activates mid-sequence	Sequence stops; the next run excludes it	Injected: mark a row Activated	Pass

#	Type	Scenario	Expected behaviour	How to test	Result
17	Edge	Merchant replies angry, or asks why closed	Sequence stops, a person answers. R10: no reason beyond category. R12: Compliance answers	Injected reply	Designed
18	Edge	Merchant replies STOP	Opt-out recorded, no further messages	Injected reply	Pass (reply box in the app's Outbox tab)
19	Edge	Two merchants tie on score	Lower application ID ranks first, same order every run	Injected tie	Pass
20	Edge	Call budget raised from 10 to 12	A3054 and A3050 are added, nothing else moves	Change the setting	Pass
21	Edge	Regional message runs over 1,000 characters	Blocked; library wording shortened	All 1,272 variants checked	Pass after trimming
22	Edge	Compliance clears an R12 case	Re-enters the pipeline and is diagnosed from the start	Injected status change	Designed
23	Data anomaly	Idle longer than account age (21 rows, for example A3016)	"Date check" flag, decision unchanged, raised with the data owner	Pack	Pass
24	Data anomaly	Duplicate application ID	Run stops with an error, nothing drafted	Injected	Pass
25	Data anomaly	Missing or unsupported language	English draft, amber flag	Injected	Designed
26	Data anomaly	Missing opt-in value	Treated as No: email	Injected	Designed
27	Data anomaly	Code appears at an unexpected step	Data flag, decision unchanged	Pack check: 0 of 45	Pass

Rows marked Designed are specified here but not yet built in the app. They are not claimed as passing yet.

14. Outputs

Every run produces results a person can see at once: a loud, colour-coded run report, a work queue sorted by what needs a person today, and a case card per merchant. Nothing important stays inside the code.

14.1 Run report (printed at the end of every run, colour-coded)

```
RAZORPAY ONBOARDING RECOVERY: RUN ON SNAPSHOT 30 Sep 2026   output hash d9e1209e
100 read | 25 excluded (activated) | 75 in scope
RECOVERABLE   67 applications | Rs 933.9 lakh a month
NOT RECOVERABLE  8  (5 closures, 3 Compliance)

GREEN  56  send automatically after the check
AMBER  11  NEEDS YOU: 8 calls, 3 codes to confirm
RED    8  PERSON OWNS: 5 closures to approve, 3 Compliance holds, 0 human review
BLOCKED 0  guardrail failures
DATA   21  date-check flags (idle longer than account age)

TOP CAUSES  name mismatch 16 (Rs 132.2 lakh) | policy pages 7 (Rs 128.1 lakh) | GST 6 (Rs 106.3 lakh) |
unknown, abandoned 22 (Rs 307.6 lakh)
CALLS THIS WEEK  A3031 A3081* A3033 A3088 A3018 A3022* A3068 A3023 | stuck small: A3009 A3020
* confirm the inferred code first
```

14.2 Output table (one row per application, the brief's table plus extras)

- **Brief columns:** application ID, diagnosed cause, fixable, recoverable, drafted message in the merchant's language, channel.
- **Decision columns:** evidence (catalog, inferred, unknown), action, escalation team, risk level, light, lane, call slot, send status and the reason for any hold.
- **Message columns:** language, email subject, Day 0 message, Day 3 and Day 7 follow-ups, English meaning, guardrail result.
- **Context columns:** priority rank and score, days idle and band, attempts, support contacts, TPV, step, business type, data flag.
- Rows are coloured by light and sorted by priority rank; filtering the light column gives each queue. Filters: light, lane, language, owner, code.

14.3 RM work queue ("Needs you today")

One list per owner, showing only amber and red cases, oldest first within each light. Each row has the merchant ID, light, what to do in one line ("Call: GST mismatch, Tamil, 4 attempts"), and the time left before escalation. Search by application ID opens the case card.

14.4 Case card (one screen per merchant)

- Header: application ID, light, lane, call slot, TPV, days idle, language, channel.
- What is wrong, in plain English, with the evidence (code, or the reviewer text it was inferred from).
- What to do: the fix, and the call script ("Open with the problem and fix, confirm they have the written message, ask what is blocking them. Never mention screening, risk or approval odds.")
- The message in the merchant's language with the English meaning beside it, plus the Day 3 and Day 7 follow-ups.
- Decision receipt: each rule that fired, in order, with the library version and run hash.
- Buttons: approve, pause, mark called, confirm code, swap a library piece, override with a reason.

14.5 Product team report (weekly)

Fixable causes ranked by number of merchants and by TPV, with the week-on-week change and the share of abandoned cases answering "What stopped you?".

Top causes, two views. The summary groups R01 (PAN name) and R03 (bank account name) as one name-mismatch cause, because one product fix covers both. The table also shows the two codes apart, so the ranking can be checked either way. Ranked by TPV alone with the codes split, the top three would be website policy pages (R06), signatory ID (R09) and GST (R02). Fixable causes total 45 applications and Rs 626.3 lakh a month; with the 22 abandoned cases that is the 67 recoverable and Rs 933.9 lakh.

Cause	Codes	Applications	TPV (Rs lakh a month)	Rank by applications, by TPV
Name mismatch, grouped (summary view)	R01 + R03	16	132.2	1, 1
Website policy pages missing	R06	7	128.1	2 (tie), 1
Signatory ID missing or mismatched	R09	3	112.1	7 (tie), 2
GST certificate invalid or mismatched	R02	6	106.3	4, 3
Document unreadable	R05	4	86.5	5 (tie), 4
Bank account name mismatch	R03	7	77.3	2 (tie), 5
PAN name differs from business name	R01	9	54.9	1, 6
Bank verification failed	R04	4	40.7	5 (tie), 7
Website not live	R07	3	15.1	7 (tie), 8
Declared category differs from website	R11	1	3.3	9 (tie), 9
Address proof missing or outdated	R08	1	2.0	9 (tie), 10
Abandoned, reason unknown (not ranked)	R13	22	307.6	Cause unknown

First product fix to ship (recommendation). Show the name found on the PAN or bank account next to the business name at the moment the merchant uploads or enters it, with a one-tap choice to correct either one. It targets the largest grouped cause (16 applications, Rs 132.2 lakh a month) and removes the mismatch before a reviewer rejects the application. Next in line: a pre-submit check for the four website policy pages (7 applications, Rs 128.1 lakh, the largest single code by TPV). How we would know it worked: fewer name-mismatch rejections per 100 applications, and a shorter time from sign-up to activation. The 22 abandoned cases (Rs 307.6 lakh) are the largest block of all, but their cause is unknown; the one-tap "What stopped you?" question in section 12 is how that gap closes.

14.6 Send log (dry run in this build)

What would be sent, to whom, on which channel and on which day, so the sequence can be checked before anything is live.

Files in this build: the output workbook (Summary, Output_75, Excluded_25, Message_Library, Sequence_Rules), the agent script, the message library and the test scripts.

15. Success criteria

Hard gates must all pass before anything is sent; a single failure stops the release. Quality targets measure how good the diagnosis and messages are. Live metrics are what we would track once real merchants are messaged; no recovery rate is claimed today.

Type	Criterion	Target	Today
Hard gate	Messages with a promise, date, amount or compliance reason	0	0 of 75 rows; 0 of 1,272 variants
Hard gate	Nudges sent to R10, R12, unclassified or Rejected cases	0	0
Hard gate	WhatsApp messages without opt-in	0	0
Hard gate	Closures sent without a person's approval	0	0 (all 5 held)
Hard gate	In-scope rows with a cause, fixable, recoverable, action and a message or a reason for none	75 of 75	75 of 75
Hard gate	Same output on two runs (output hash)	Identical	Identical
Hard gate	Real or confidential data used	None	None
Quality	Diagnosis agreement with hand labels written before the run (30-case gold set)	80% or more, every miss explained	Pending: labels first
Quality	Safety routes (R10, R12) correct in the gold set	100%	Pending
Quality	Regional library signed off by native speakers before any regional send	100%	Done
Quality	WhatsApp messages within 1,000 characters in every language	100%	100% (longest 823)
Quality	Amber cases cleared within their time limit	95%	Measured live
Live metric	Resume rate within the sequence, by code, lane and language	Baseline in weeks 1 to 2, then improve	Not measurable on the pack
Live metric	Activated TPV recovered per week	Track weekly	Not measurable on the pack
Live metric	Reply rate, opt-out rate, complaint rate	Opt-out and complaints kept low; any rise pauses the sequence	Not measurable on the pack
Live metric	Human override rate and guardrail block rate	Falling over time	Not measurable on the pack
Live metric	Share of abandoned merchants answering "What stopped you?"	Rising	Not measurable on the pack

Type	Criterion	Target	Today
Live metric	Calls versus no calls: resume rate of called stuck small merchants against similar uncalled ones	Decides whether calls are worth the time	Not measurable on the pack

What good could look like (illustrative, not a forecast). The pack has no outcome data, so these figures are arithmetic, not predictions: the Rs 933.9 lakh cohort at three assumed resume rates, assuming resumed merchants carry the average TPV. A live pilot replaces them with a measured baseline in weeks 1 to 2.

Assumed resume rate	Applications back on track (of 67)	Monthly TPV unlocked (Rs lakh)
10%	7	93.4
20%	13	186.8
30%	20	280.2

16. Test plan

Each part is tested on its own, then the whole agent runs on all 100 rows. Results are reported as they are, including what failed.

#	Test	What it checks	Pass rule	Result today
1	Input checks	Duplicates, missing fields, unknown languages, codes at the wrong step, date consistency	Run stops on a duplicate; everything else flagged	Pass: an injected duplicate stops the run; 0 of 45 codes at a wrong step; 21 date flags raised
2	Classification and adversarial tests	Every code, Hinglish text, blank codes, safety override, unrecognised text, bad messages	37 of 37	37 of 37
3	Message variants	Every code, tone, channel, empathy state and follow-up, in six languages; plus the name fallback, steps-left line, HELP and STOP replies, 1,000-character limit and no emoji (16 app checks)	0 checker failures	1,272 checked, 0 failures (936 Day 0 and 336 Day 3 and Day 7 variants, WhatsApp and email), re-run after the 7 Nov message format change; 16 of 16 app checks pass
4	Full dry run	All 100 rows end to end	75 in scope, 25 excluded, every in-scope row has an outcome, 0 failures	Pass
5	Reproducibility	Two runs on the same input	Same output hash	Pass
6	Injected edge cases	Rejected with a fixable code, activated mid-sequence, screening text on a fixable code	Behaviour as in section 13	Pass after one fix
7	Gold set	30 cases hand-labelled by the owner before the agent is run: all 4 inferred cases, all 8 closure and Compliance cases, and 19 random others. Cause and action are compared with the system. A self-check, not an independent score: the organisers' hidden answers are the independent test	80% or more on cause, 100% on safety routes; every disagreement listed	Pending
8	Native-speaker review	Tamil, Kannada and Marathi library read by native speakers	Signed off, or corrections applied	Done
9	Brand and tone review	English library against the tone-of-voice guidelines	Owner approval	Pass

#	Test	What it checks	Pass rule	Result today
10	Demo test send (optional)	Three messages sent to the builder's own inbox, never to a merchant	Arrive intact, correct format	Optional

17. Risks and controls

The highest risks are revealing a compliance reason, helping a restricted business around the rules, and a wrong or badly translated message; each has at least two controls.

Risk	Area	Impact	Controls	Left over
Compliance reason revealed to a merchant	Compliance	High	Fixed holding note only; screening words blocked in six languages; callers' script forbids it; Compliance approves wording	Low
Workaround offered to a restricted business	Compliance	High	Closure has no link or next step; workaround words blocked; a person approves each closure	Low
Wrong diagnosis sends the wrong fix	Wrong answer	Medium	Catalog code first; inferred codes need AI and rules to agree plus a person; safety override; gold-set test	Medium until the gold set is done
AI misreads blank-code text	Wrong answer	Medium	AI only picks from the catalog list, needs 80% confidence and agreement with rules, cannot pick a route or write text; a person confirms	Low
Reviewer text contains instructions aimed at the AI	Security	Low	AI output is limited to one code and a number; anything else is ignored and sent to a person	Low
Bad Tamil, Kannada or Marathi wording	Language	Medium	Approved by native speakers; English meaning beside each message; script check	Low
Promise or date slips into a message	Wrong answer	High	No free text; digits and promise words blocked	Low; vague promises are the known gap
WhatsApp without consent, or too many messages	Privacy, trust	Medium	Opt-in check; at most 3 messages; STOP line; 9am to 7pm window; stop on reply	Low
Personal data exposed in a message	Privacy	Medium	No PAN, account numbers, names, dates or amounts in text; resume link unique per application	Low
Ranking ignores small merchants	Bias	Medium	Score only rations calls; 2 stuck small-merchant slots; fairness view; everyone gets the sequence	Low
Thresholds create cliffs (29.9 versus 30 lakh)	Bias	Low	Calls go by score, not by the lane line	Low
Bad input data	Data	Medium	Input checks; date flags; duplicates stop the run	Low

Risk	Area	Impact	Controls	Left over
Amber queue piles up	Operations	Medium	Time limits with escalation; kill switch; budget is a setting	Medium
Results cannot be reproduced	Deal-breaker	High	Fixed rules, saved AI answers, ID tie-break, output hash	Low
Overclaiming in the pitch	Credibility	High	No recovery rate claimed; designed versus built marked; failures listed	Low

18. Assumptions and open questions

Every number we chose ourselves is listed here, with how it would be checked. The brief says to state an assumption clearly and move on, and these are the ones.

Assumptions

Assumption	Why we chose it	How to check it
Snapshot date is 30 Sep 2026	Stated in the Data Pack README	Not needed
R12 is not recoverable until Compliance clears it	The catalog says "needs compliance review" and forbids discussing it	Confirm with Compliance
Every merchant has an email from sign-up	The pack has no contact columns	Data owner
Each application has a unique resume link	Needed for one-tap return	Product team
Call budget of 10 a week, split 8 plus 2	A small team; a setting the ops lead can change	Ops capacity
Lane A at TPV of 30 lakh or more	A clear cut that captures 35% of recoverable TPV with 7 merchants	Review after a month
"Stuck" means 3 or more attempts, or 2 or more support contacts	Repeated effort without success	Compare resume rates
Freshness, ease and evidence weights	Business logic; robust: other weights give the same top 10	Refit on real resume rates
Day 0, 3 and 7 cadence; 9am to 7pm window	Common practice, not too pushy	A/B test the timing
Merchants idle 31 days or more get one message only	More messages to a cold merchant feel like spam	Measure replies
days_since_last_activity is trusted despite 21 date anomalies	Changing it would invent data	Data owner
AI confidence floor of 80%	Small volume, high cost of a wrong fix	Gold set
Amber time limits (same day, 1 week, 2 days)	Nothing should wait forever	Ops lead
Steps run in this order: Business details, KYC documents, Bank verification, Website/App review; step_reached is the step the merchant is on	The data pack names the steps but not their order	Product team
WhatsApp messages stay under 1,000 characters	Our own limit; WhatsApp template bodies allow about 1,024	Check with the WhatsApp provider
A reply of HELP goes to a person	The message tells the merchant our team will help	Onboarding Ops

Assumption	Why we chose it	How to check it
The P3 data has no merchant name, so everyone gets "Dear Valued Merchant"	A name is used only if the data holds one that passes the checks	Data owner

Open questions (owner and status)

- ☐ Compliance confirms the holding-note wording and the pause rule.
- ☑ Native speakers reviewed and approved the Tamil, Kannada and Marathi library.
- ☐ Owner confirms the Hinglish and Hindi library.
- ☐ Data owner explains the 21 date anomalies.
- ☐ Hand-label the 30-case gold set before the agent is run, then compare.
- ☑ Build platform chosen: Streamlit Community Cloud with the Python app, deployed from GitHub (same code, identical output on every run).

19. Tools and AI in the build

The agent runs on fixed rules and one narrow AI step, so the same input always gives the same output.

Part	What to use	Rule
Platform	Streamlit Community Cloud with the Python app, deployed from GitHub	Any tool that gives identical output on every run
Data handling	pandas for the Data Pack; openpyxl for the output workbook	No network calls except the AI step below
Message text	The approved library: Appendix A (English) and the Message_Library in the output workbook (six languages)	No other wording is ever sent
Rules	Keyword lists per code, forbidden-word lists per language and script, thresholds from sections 6 and 12	Kept in one rules file so a change is reviewable
AI step (runtime)	One model call per application with a blank code and reviewer text (4 in the pack)	See below
Build assistance	Claude (chat and Claude Code) helped draft the code, the message wording and this document	The author reviewed every rule, threshold and message and is accountable for each

The AI step, exactly

- **Input:** the reviewer text, plus the catalog codes R01 to R12 with their plain meanings.
- **Output, in a fixed format only:** one code or UNCLEAR, a confidence from 0 to 100, and the exact words relied on. Anything else counts as UNCLEAR.
- **Settings:** temperature 0. The answer is saved in a file keyed by application ID and the text, and every rerun reads the saved answer.
- **Use:** combined with the keyword rules as in section 7. Below 80%, a disagreement, or any R10 or R12 answer goes to a person.
- **Never:** choose a route, close an application, contact Compliance, pick a call slot or write any text a merchant sees.

Why only one AI step. A catalog code is present for 49 of 75 applications, and rules read it exactly, the same way, every time. AI helps only where rules cannot read free text, which is 4 applications here. Keeping it this narrow is what makes every run repeatable and every decision explainable.

Production-readiness additions (sections 20 to 30)

This part adds requirements that were not fully specified in sections 1 to 19. These additions are proposed controls for a production implementation. They do not change the data-pack results or claim that the controls have already been built or tested.

20. Requirement status and release scope

Every capability must carry one implementation state in product tracking and release reporting:

- Built: implemented and demonstrated in the current prototype. This is not equivalent to production-ready.
- Designed: behaviour is specified but implementation and verification remain outstanding.
- Proposed: a candidate enhancement that is not required for the first controlled release.
- Production-ready: implemented in the target environment, security-reviewed, operationally owned, tested against approved acceptance criteria, and explicitly signed off.

No live communication may be enabled solely because a feature is marked Built in the prototype. The dry-run status remains the release baseline until all gates in Section 28 are evidenced.

21. System context and integration contracts

The production architecture must preserve the principle in sections 6 and 7: deterministic business rules own routing; the narrow AI step may only suggest a catalog code for blank-code text. The AI component must not decide eligibility, compliance disposition, closure, message content, channel, priority, or send timing.

21.1 Logical components

- Application data adapter: reads the approved onboarding application snapshot or event feed and validates schema, freshness, completeness and source identity.
- Catalog and policy service: provides versioned rejection-code definitions, fixability, standard fixes, restricted-category rules and Compliance routing rules.
- Deterministic decision engine: applies status gates, safety overrides, recoverability rules, lane assignment, priority score and case ownership.
- Classification assist: invokes the constrained AI classification step only for eligible blank-code records; records model, prompt, catalog and output versions.
- Message assembly and guardrail service: composes only from approved, versioned message-library components and runs language-specific safety checks.
- Orchestration and delivery adapter: schedules, dispatches, retries and reconciles approved communications through configured providers.
- Case management and audit store: records state transitions, evidence, approvals, overrides, communication events and decision receipts.
- Reporting layer: exposes queue, funnel, cause, TPV-at-stake, delivery and outcome metrics with access controls.

21.2 Minimum input contract

Field group	Required contract	Failure behaviour
Identity	application_id unique and immutable; source_system; snapshot/event timestamp; schema version	Duplicate or missing application_id stops the batch. Preserve source record and error reason.

Field group	Required contract	Failure behaviour
Application state	application_status; current onboarding step; last activity timestamp or validated days_since_last_activity	Unknown status or stale state blocks automated action and enters data-quality review.
Diagnosis	rejection_code; rejection_message; catalog_version	Unknown code or catalog mismatch: no nudge; route to human triage.
Contact and consent	verified email/phone destination; channel eligibility; WhatsApp opt-in evidence, timestamp and source; opt-out/suppression status	Missing or ambiguous consent blocks WhatsApp. Do not assume that an email exists or is valid; block dispatch until a verified eligible channel exists.
Personalisation	preferred_language; approved language-library version	Unsupported or unapproved language: no automated regional send; route to approved fallback policy and human review.
Recovery action	resume-link identifier, application binding, expiry, destination domain and status	Missing, expired, non-unique or mismatched link blocks message dispatch.
Commercial context	expected_monthly_tpv_inr_lakh and currency/period definition	Missing or invalid TPV excludes call scoring and is flagged; it must not change safety routing.

21.3 Minimum output contract

For every application, persist: application_id; run_id; snapshot_id; decision timestamp; status; diagnosed cause; evidence source; fixable and recoverable flags; route; risk light; owner; lane; priority score and factor values; message-library version; language; channel; guardrail result; delivery eligibility; next action; hold reason; and immutable decision receipt.

The PRD's current assumptions about a sign-up contact and a unique resume link must be converted into verified upstream contracts before production. An assumption is not an integration guarantee.

22. Case lifecycle and communication orchestration

Implement an explicit state machine. State changes must be event-driven, idempotent and auditable. A case may have only one active recovery sequence at a time.

State	Entry condition	Permitted action / transition
RECEIVED	Application snapshot/event accepted	Validate schema, identity, freshness and required fields.
VALIDATION_HOLD	Input or contract validation fails	No draft or send. Assign data owner and record blocking reason.
DIAGNOSING	Validated, non-activated application	Apply deterministic code/catalog rules and constrained classification assist where permitted.
HUMAN_REVIEW	Ambiguous diagnosis, amber case, inferred code, or explicit review gate	Human confirms, corrects, pauses or escalates with reason.

State	Entry condition	Permitted action / transition
COMPLIANCE_HOLD	R12 or screening signal	Compliance owns case; only approved holding communication may be considered.
CLOSURE_PENDING_APPROVAL	R10/restricted category route	Ops lead reviews each case; no automatic closure or merchant message before approval.
ELIGIBLE_TO_MESSAGE	Recoverable case, approved library, verified channel and valid link	Final pre-send checks run; create an idempotent send request.
SCHEDULED	Message approved and scheduled within contact window	Cancel or suppress on activation, reply, opt-out, compliance hold, case change or kill switch.
SENT_PENDING_DELIVERY	Provider accepted request	Reconcile provider delivery status; do not equate API acceptance with delivery.
DELIVERED	Provider confirms delivery	Start follow-up timer from the defined delivery event, not merely queue creation.
DELIVERY_FAILED	Provider reports permanent failure or retry policy exhausted	Stop sequence; route to Ops; do not switch channel without valid eligibility and policy.
REPLIED	Any inbound response received	Cancel pending follow-ups; route thread to a human. No AI-generated reply in this release.
ACTIVATED	Authoritative onboarding state indicates activation	Cancel all future communications and close recovery sequence.
OPTED_OUT	STOP or equivalent suppression event	Record suppression and cancel all future messages across applicable channels.
CLOSED	Approved closure or final non-recovery outcome	No further recovery nudges unless a separately approved re-entry event occurs.

22.1 Final pre-dispatch gate

Immediately before each dispatch, re-read or validate the authoritative application status, consent/opt-out state, compliance flags, contact destination, resume-link validity, message-library approval, active-sequence count, send-window eligibility and global/per-case kill-switch state. If any check is unavailable, stale or fails, fail closed: do not send.

22.2 Delivery and retry policy

- Persist a unique idempotency key per application, sequence step, channel and message version. Provider retries must not create duplicate merchant messages.
- Store provider message ID, request timestamp, acceptance status, delivery status, failure code, retry count and final disposition.
- Separate transient failures from permanent failures. Retry only transient failures under an approved bounded policy; never retry after opt-out, reply, activation, compliance hold or kill switch.

- Do not silently fall back from WhatsApp to email. Channel fallback requires a verified address, a permitted contact basis and an auditable policy decision.
- Define the follow-up clock explicitly. Section 12 sets Day 0, 3 and 7; product and Ops must decide whether that clock starts at approval, dispatch or confirmed delivery. Recommended control: start from confirmed delivery.
- Confirm the R12 holding-note cadence (Day 0 and Day 7, section 12) with Compliance and record one approved cadence before implementation.

23. AI classification, evaluation and model governance

The AI step remains a constrained classification aid. It is not an autonomous agent for merchant-facing decisions.

23.1 Classification output and controls

- Input must be limited to the minimum reviewer text and the approved catalog definitions required for classification.
- Output schema: one catalog code or UNCLEAR; confidence; exact supporting text span; model identifier; prompt version; catalog version. Reject malformed, extra or ungrounded output.
- Treat reviewer text as untrusted data. Ignore embedded instructions, links, requests to reveal prompts, or attempts to alter system behaviour.
- R10 and R12 predictions, any screening/restricted-category indicators, low confidence, disagreement, unsupported language or missing evidence always route to a human; no merchant message is drafted from an unresolved classification.
- Retain the deterministic rule result and AI suggestion separately. A human correction must not overwrite the original model output.

23.2 Evaluation requirements

Evaluation dimension	Required measurement	Release expectation
Classification quality	Precision, recall, F1 and confusion matrix by code; report macro and weighted metrics	Thresholds approved per code based on risk; aggregate accuracy alone is insufficient.
Safety-route detection	False-negative count and rate for R10, R12, restricted-category and screening signals	Zero tolerated false negatives in the approved release test set.
Language robustness	Performance by language/script, code-switching, spelling variation and transliteration	No automated classification for unsupported or unvalidated language.
Abstention	UNCLEAR rate and accuracy among non-abstained cases	Abstention is preferred to a wrong high-risk classification.
Stability	Repeatability across repeated runs and model/prompt upgrades	Regression suite passes before any version change.
Human agreement	Blind-labelled gold set, adjudication process and disagreement analysis	The existing 30-case / 80% target is prototype evidence only; define a statistically and operationally justified production set and per-risk thresholds.

Evaluation dimension	Required measurement	Release expectation
Adversarial resilience	Prompt injection, irrelevant text, contradictory evidence and malicious payloads	No policy bypass, no free-text output, no unsafe route.

23.3 Model and prompt change management

Pin model version, prompt version, inference parameters, schema version and catalog version. Every change requires offline regression, safety-route validation, language review where affected, approval by the product owner and rollback capability. If the provider changes a model behind a stable name, treat it as a version change and revalidate.

24. Security, privacy and data governance

- Define a data inventory and purpose limitation for every field. Do not send PAN, bank-account numbers, identity documents, full reviewer notes or unnecessary personal data to the model or messaging provider.
- Encrypt data in transit and at rest; manage keys through the approved secrets/key-management service. Never place API keys, credentials or personal data in code, logs or test fixtures.
- Apply role-based access control: Ops agent, Ops lead, Compliance, language reviewer, product analyst, platform administrator and auditor. Enforce least privilege and separation of duties for closure and Compliance approvals.
- Mask or redact personal data in dashboards, logs, exports and support tooling. Restrict bulk export and record access/export events.
- Use short-lived, single-application resume links with unguessable tokens, expiry, revocation, TLS and server-side authorisation. Do not expose application identifiers or personal data in URLs.
- Define retention and deletion periods for source snapshots, model inputs/outputs, message content, provider logs, audit events and analytics extracts. Retention must be approved by Security, Privacy and Legal.
- Record consent evidence and opt-out suppression centrally. Honour opt-out across channels and prevent re-imported snapshots from re-enabling contact.
- Complete vendor/privacy review for AI and messaging providers, including data residency, subprocessors, training-use terms, incident notification, deletion and access controls.
- Security review must include threat modelling for prompt injection, broken object-level authorisation, link enumeration, replay, duplicate dispatch, webhook spoofing and privilege escalation.

25. Operational ownership, SLAs and incident response

Queue / event	Owner	Required operating rule
Amber inferred-code review	Onboarding Ops	Same-day review as set in section 11; unresolved cases remain blocked.
Regional-language approval	Named native-speaker reviewer / language owner	No send in a language until the relevant library version is signed off.
R10 closure approval	Onboarding Ops lead	Case-by-case approval with reason and audit receipt.
R12 / screening hold	Compliance	Compliance owns wording, cadence, pause and case disposition.

Queue / event	Owner	Required operating rule
Data contract failure	Named data owner	Batch or record held according to severity; correction must preserve source lineage.
Provider delivery failure	Messaging operations / platform owner	Reconcile status, apply bounded retries and route permanent failures.
Security/privacy incident	Security incident owner	Disable affected integration or all sends, preserve evidence and follow incident process.
Global or scoped kill switch	On-call operations owner	Document who can activate, who can deactivate and mandatory approval for restart.

25.1 Incident playbooks

- Wrong or unsafe message: stop all affected sends, identify impacted applications and message versions, notify Ops/Compliance/Privacy as required, preserve audit evidence, and resume only after root cause and regression fix are approved.
- Consent or opt-out breach: disable the affected channel, suppress impacted recipients, reconcile provider logs and complete incident review before restart.
- Incorrect R10/R12 routing: stop the full workflow, route all potentially affected records to Compliance, and reprocess only after approved rule correction.
- Duplicate or out-of-window dispatch: disable dispatch, reconcile idempotency keys and provider events, correct scheduler/clock logic and verify with replay tests.
- AI/model regression: revert to the last approved model/prompt or disable AI classification; unresolved cases go to human review.

26. Product analytics and experiment design

Separate operational throughput, communication delivery, merchant behaviour and business outcomes. The synthetic pack can establish cohort size and TPV at stake; it cannot establish incremental recovery or realised TPV.

Metric	Definition	Required breakdown / notes
Eligible cohort	Unique applications meeting approved status and policy gates in the measurement period	Status, code, language, channel, lane and snapshot date.
Message delivery rate	Provider-confirmed delivered messages / provider-accepted eligible messages	Report separately by channel and provider; exclude queued/accepted from delivered.
Merchant response rate	Unique eligible applications with an inbound response within the defined window / delivered applications	Classify response intent only after human review in the first release.
Resume rate	Applications with authoritative onboarding progress after communication / delivered eligible applications	Define qualifying progress event and attribution window.

Metric	Definition	Required breakdown / notes
Activation rate	Applications reaching Activated status / eligible applications	Report gross observed rate and incremental lift separately.
Incremental activation lift	Treatment activation rate minus comparable control activation rate	Use randomised holdout where policy and safety permit; otherwise disclose observational limitations.
Activated TPV	Sum of the approved TPV measure for applications that activate	Do not label all activated TPV as caused by the agent; distinguish observed and incremental.
Incremental TPV	Estimated treatment-control difference in activated TPV under the approved experiment design	Pre-register methodology, window, exclusions and uncertainty.
Opt-out rate	Unique opt-outs / delivered messages, by channel	Define denominator and deduplication; pause on approved threshold breach.
Complaint rate	Unique complaints / delivered messages	Include provider complaints and internally received complaints.
Human override rate	Cases where an authorised human changes diagnosis, route, message or disposition / reviewed cases	Segment by reason and reviewer; use for rule improvement.
Guardrail block rate	Messages blocked by guardrails / messages evaluated	A rising rate triggers investigation; never optimise by suppressing valid blocks.

26.1 Controlled pilot

- Start with a dry-run shadow period using production-shaped data and no merchant dispatch.
- Proceed to a limited, explicitly approved pilot only after all hard gates pass. Use a holdout/control group where operationally and ethically appropriate.
- Predefine eligible cohort, exclusions, randomisation unit, duration, outcome window, primary metric, guardrail metrics and stop conditions before launch.
- Do not claim the two called small-merchant slots demonstrate causal lift without a suitable control design and adequate sample size.
- Report results by cause, language, channel, business type and merchant-size cohort to detect uneven outcomes.

27. Acceptance criteria and traceability

The following criteria supplement the existing prototype test plan. Each must be linked to an implementation ticket, test case, evidence artifact and named approver.

ID	Acceptance criterion	Evidence required
AC-01	Every input row receives exactly one terminal or active case state; no application is silently dropped or duplicated.	Reconciliation report: input count = excluded + in-scope outcomes; unique application IDs.

ID	Acceptance criterion	Evidence required
AC-02	Activated applications are excluded and never receive recovery communications, including when activation occurs after scheduling.	State-transition and pre-dispatch race-condition tests.
AC-03	R10, R12 and any restricted/screening signal cannot enter the automated fix-it path.	Gold-set and adversarial tests; zero safety-route false negatives.
AC-04	No WhatsApp message is dispatched without valid, current, auditable opt-in; no message is sent after opt-out.	Consent and suppression integration tests plus provider evidence.
AC-05	No message is dispatched without a verified destination, valid application-bound resume link and approved language-library version.	Negative tests for missing, expired, invalid and mismatched values.
AC-06	All dispatches pass a final authoritative status, consent, compliance, time-window and kill-switch check.	Pre-dispatch integration and race-condition tests.
AC-07	Retries and replay do not create duplicate messages; provider delivery states reconcile to the audit store.	Idempotency, webhook-signature and replay test evidence.
AC-08	Every decision and human override is traceable to input snapshot, rules/catalog version, model/prompt version where applicable, library version and actor.	Sample audit receipts and access-control tests.
AC-09	Only approved template components can appear in merchant-facing content; prohibited promises, dates and compliance disclosures are blocked.	Six-language guardrail suite and human review evidence.
AC-10	All production roles enforce least privilege and separation of duties for closure and Compliance actions.	RBAC matrix, negative permission tests and security sign-off.
AC-11	Dashboards distinguish estimated TPV at stake, observed activated TPV and incremental TPV.	Metric dictionary, dashboard validation and analytics sign-off.
AC-12	Rollback, kill switch, provider outage and incident procedures are executable by named owners.	Operational exercise / tabletop evidence and runbook approval.

28. Production release gates

Live merchant communication is blocked until every mandatory gate below is complete. A partial pass is not approval.

Gate	Required evidence	Approver
Policy and route safety	Compliance-approved R10/R12 behaviour, restricted-category overrides, holding-note cadence and zero safety-route misses in release tests.	Compliance + Product
Data and integrations	Signed input/output contracts, authoritative status source, verified contact/consent, valid resume-link service and reconciliation.	Engineering + Data owner
Language readiness	All six language libraries signed off for the current version (Tamil, Kannada and Marathi approved by native speakers). Any new version needs sign-off before send.	Language owner + Ops
AI quality	Gold set, per-code metrics, false-negative analysis, adversarial suite, pinned model/prompt and rollback tested.	Product + AI owner + Compliance
Security and privacy	Threat model, RBAC, encryption, retention, vendor review, consent/opt-out and link-security sign-off.	Security + Privacy/Legal
Messaging operations	Provider delivery webhooks, idempotency, bounded retries, time-window enforcement, suppression and kill switch tested.	Engineering + Messaging Ops
Operational readiness	Named queue owners, SLAs, escalation paths, incident playbooks, support training and on-call coverage.	Onboarding Ops lead
Measurement	Metric dictionary, baseline, experiment design, guardrail thresholds and reporting validated.	Product Analytics + Product
Final launch approval	All above evidence linked to release record; dry-run reconciliation passes; pilot scope and rollback owner recorded.	Product owner + designated business approver

29. Decision and dependency register

The following decisions are left open in sections 1 to 19 and must be closed before the relevant release gate can pass.

Decision / dependency	Why it matters	Owner to assign	Status
Confirm source-of-truth application status and event freshness	Prevents messages after activation or a status change.	Engineering / Onboarding platform	Open
Confirm contact data, verification and consent evidence	The synthetic pack has no phone/email destination; assumptions cannot authorise contact.	Data owner + Privacy	Open
Confirm resume-link creation, expiry, revocation and application binding	A placeholder link cannot be used in production.	Onboarding platform	Open
Resolve R12 cadence and exact holding-note wording	Section 12 sets Day 0 and Day 7, but Compliance must sign off the cadence and the wording.	Compliance	Open
Approve language library and fallback behaviour	Regional messages must not be sent before native-speaker sign-off.	Language owner	Done (current version)
Define production AI evaluation thresholds and gold-set size	Prototype 30-case / 80% target is insufficient as a production safety specification.	Product + AI owner + Compliance	Open
Define delivery retry, webhook and channel fallback policy	Avoid duplicates, silent channel switching and false delivery assumptions.	Messaging Engineering	Open
Approve retention, deletion, vendor and data residency controls	Required for personal data and external AI/messaging processing.	Security + Privacy/Legal	Open
Approve pilot design, baseline and stop thresholds	Separates correlation from incremental recovery impact.	Product Analytics + Ops	Open
Select and approve production platform and ownership model	The prototype runs on Streamlit; the production platform and ownership are not chosen.	Engineering leadership	Open

30. Change log

Revision	Change	Treatment
1.2	Added document navigation, consistent formatting, production integration contracts, case lifecycle, delivery controls, AI evaluation, security/privacy, operations, metrics, acceptance criteria, release gates and decision register.	Proposed additions; require owner review and approval.
1.2	Reworded executive summary to distinguish estimated TPV at stake from realised or incremental recovered TPV.	Clarification based on data limitations.
1.2	Clarified that prototype Built status does not mean production-ready or live-send approved.	Release governance clarification.
1.2	Flagged R12 holding-note cadence as requiring a single Compliance-approved decision.	Open decision for Compliance.
1.2	Brought sections 1 to 19 in line with the 7 November build: 1,000-character limit, steps-left and HELP format, 1,272 variants, output hash c36ea57c, AI second opinion marked built.	Matches the current app.
1.2	Added the two views of the top causes, the first product fix, illustrative resume-rate scenarios and hand-labelled gold-set wording.	Review improvements.
1.2	Removed the second cover page, split index and placeholder text; one index now lists every section; sub-headings and tables formatted consistently.	Formatting clean-up.
1.3	Native speakers approved the Tamil, Kannada and Marathi messages, so the regional review hold is removed (56 automatic, 11 check first, 8 person handles). Added email subjects for Day 3 and Day 7. Output hash is now d9e1209e. New app with a case-first view and a product-team page.	Matches the new app.

Figures are from the synthetic data pack and the prototype run on 7 November 2026. New production controls are recommendations and must not be represented as already implemented.

Appendix

Appendix A: Message library (English master)

These are the approved English pieces (v2 format, 7 Nov); every message is assembled from them. The shared lines (greeting, openers, steps-left line, help, STOP and sign-off) are in section 9. The other five languages are in the app's More > Message library tab and the Excel download, one piece per row. That library is the single source of truth for non-English text; Tamil, Kannada and Marathi rows were reviewed and approved by native speakers.

Code	Short issue (follow-ups)	Problem line (bold clause marked)	Fix bullets
R01	the PAN name match	We noticed that **the name on your PAN card does not match the business name in your application.**	(choose one) • Upload the PAN of the business (or of the owner, for a proprietorship) that matches your application • Or correct the business name in your application to match your PAN
R02	your GST details	We could not validate your GST details. **The GSTIN may be wrong, inactive, or linked to a different legal name or address.**	(choose one) • Enter the correct GSTIN • Or upload a current GST certificate that matches your legal name and address
R03	the bank account name	We noticed that **the name on your bank account does not match your business or PAN name.**	(choose one) • Use a current account in the business name (the owner's name for a proprietorship) • Or upload a bank letter or cancelled cheque that shows the matching name
R04	your bank account verification	We could not verify your bank account with the small test credit.	• Re-enter the account number and IFSC carefully • Check that the account is active and can receive credits
R05	a clearer document (Day 7: uploading a clearer document)	We could not read your document. **It may be blurred, cropped or password-protected.**	• Upload a clear, full-page colour scan • Use a PDF, JPG or PNG file under 5 MB
R06	your website policy pages	We noticed that **your website or app is missing some required pages.**	• Add these pages with working links: Terms & Conditions, Privacy Policy, Refund/Cancellation Policy and Contact details • Then submit your website link again
R07	your website access	We could not review your website. **It is not live, or it needs a login to see products and prices.**	• Make the website open to everyone, with products, prices and contact details visible • Then submit it again
R08	your address proof	We noticed that **your address proof is missing, out of date, or shows a different address from your application.**	• Upload a recent utility bill, rent agreement or other valid address proof • Make sure it matches the address in your application

Code	Short issue (follow-ups)	Problem line (bold clause marked)	Fix bullets
R09	the signatory ID	We noticed that **the ID of your authorised signatory or director is missing, or its details do not match your application.**	• Upload a valid ID of the authorised signatory • Make sure the name and date of birth match your application
R11	your business category	We noticed that **the business category you selected is different from what your website sells.**	(choose one) • Select the correct category • Or share a short note with evidence explaining your business model, for a second review
R13	the {step} step	We noticed you paused at **the {step} step.**	Here are your next steps: • Open the link below • Continue from the {step} step

R10 closure (sent only after a person approves): "Dear Valued Merchant, thank you for applying to Razorpay, and for the time you spent on your application. After review, we are sorry to say that we are unable to support this type of business, so we have closed your application. We truly appreciate your interest in Razorpay, and we wish you the very best for your business."

Email subjects: Day 0 "Your Razorpay setup: here is how to continue", Day 3 "A reminder about your Razorpay setup", Day 7 "Need help with your Razorpay setup?", closures and holding notes "Your Razorpay application".

The R12 holding notes and the Day 3 and Day 7 templates are in section 12.

End of document.